

CLIMATE CHANGE AND GLOBAL WARMING

Climate change refers to long-term shifts in global temperatures and weather patterns. The primary cause is human activity, particularly the burning of fossil fuels like coal, oil, and natural gas. These activities release greenhouse gases such as carbon dioxide (CO₂) and methane (CH₄) into the atmosphere.

EFFECTS OF CLIMATE CHANGE

Rising Global Temperatures: The average global temperature has increased by approximately 1.1°C since pre-industrial times. This might sound small, but it has significant impacts on ecosystems and human societies.

Melting Ice Caps: The Arctic ice cap is melting at an alarming rate. Scientists predict that summer Arctic sea ice may disappear entirely within the next 30 years.

Rising Sea Levels: As ice melts, sea levels rise. This threatens coastal cities and island nations. Some small island nations like Tuvalu and Kiribati face existential threats.

Extreme Weather Events: Climate change intensifies hurricanes, floods, droughts, and wildfires. The frequency and severity of these events have increased significantly.

SOLUTIONS TO CLIMATE CHANGE

Renewable Energy: Solar, wind, and hydroelectric power can replace fossil fuels. Countries like Denmark now generate 80% of their electricity from wind.

Energy Efficiency: Improving building insulation, LED lighting, and efficient transport can reduce energy consumption by 30%.

Reforestation: Planting trees helps absorb CO₂ from the atmosphere. One tree can absorb 25 kg of CO₂ in its lifetime.

Policy Changes: Governments can implement carbon taxes, emissions trading, and regulations to limit greenhouse gas emissions.

